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Portfolio: https://herbal-rancher.github.io/

5.0 Milestone Assessment: Multi-Story Office Building Network Infrastructure Design Proposal

Executive Summary & Business Requirements

This proposal presents the design of a secure, scalable, and highly available enterprise network infrastructure for a multi-story office environment supporting multiple business units, remote employees, centralized services, and future organizational growth. As the IT consultant for this project, my approach begins with understanding the organization's business and operational requirements and then selecting the networking technologies necessary to support them.

The proposed solution uses a hierarchical network architecture with enterprise switching technologies, VLAN segmentation, redundant infrastructure, structured IPv4 and IPv6 addressing, centralized network services, secure remote access, hybrid cloud connectivity, and layered security controls. Together, these technologies provide reliable communication, efficient data flow, simplified management, improved security, and the flexibility to accommodate changing business and technology requirements.

Successful network design begins with understanding business requirements, not simply selecting hardware. Each technology incorporated into the proposed network should address a specific business or operational need. The primary requirements guiding the design are summarized below.

Business Requirement	Design Objective
Scalability	Support additional users, devices, services, and future business growth
Security	Protect organizational systems, data, users, and network resources
Availability	Minimize downtime through resilient and redundant infrastructure
Performance	Maintain efficient traffic flow and reliable application performance
Reliability	Provide dependable connectivity and support business continuity
Manageability	Simplify configuration, monitoring, troubleshooting, and administration

Business Requirement	Design Objective
Cost Efficiency	Balance performance and resiliency with long-term operational costs

Enterprise Network Architecture

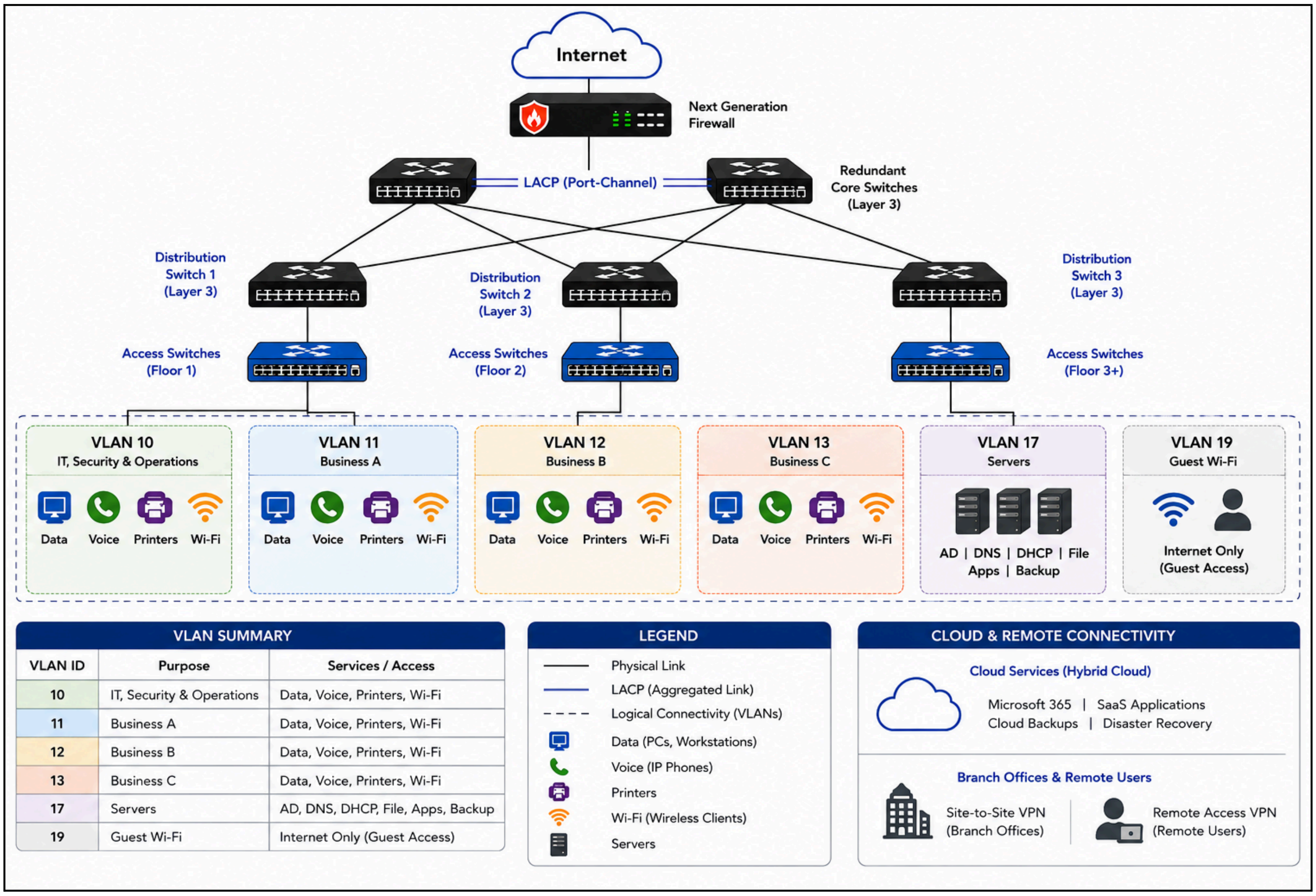

Figure 1. High-level architectural view of the proposed enterprise network illustrating hierarchical Core, Distribution, and Access layers, redundant switching infrastructure, VLAN-based logical segmentation, and scalable connectivity supporting multiple business tenants, shared enterprise services, and future organizational growth.

Switching Technologies & Features

My approach to configuring the switching infrastructure begins with the business and device requirements of each floor. Managed Layer 2 access switches will connect employee computers, IP phones, printers, wireless access points, cameras, and other endpoint devices. Redundant Layer 3 switches in the Main Distribution Frame (MDF) will provide the high-speed backbone connectivity and inter-VLAN routing. Devices will be assigned to VLANs according to their business function and security requirements, allowing the network to use the same physical switching infrastructure while maintaining separate logical broadcast domains.

To optimize performance, security, availability, and manageability, the switching infrastructure will use IEEE 802.1Q trunking to transport multiple VLANs between switches. Rapid Spanning Tree Protocol (RSTP) will be used to prevent Layer 2 loops, and Link Aggregation Control Protocol (LACP) to increase uplink bandwidth and provide redundancy. Quality of Service (QoS) will prioritize delay-sensitive voice and video traffic, while PoE+ will support devices such as IP phones and wireless access points. Port security, 802.1X authentication, Bridge Protocol Data Unit (BPDU) Guard, storm control, VLAN segmentation, and Access Control Lists (ACLs) will help prevent unauthorized access and protect the switching environment.

Feature	Purpose	Benefit
VLANs	Separate organizations and/or departments	Security & reduced broadcasts
Trunk Ports	Carry multiple VLANs	Simplifies backbone connections
802.1Q Tagging	VLAN identification	Standardized VLAN communication
802.1X Network Access Control	Authenticate users/devices before network access	Prevents unauthorized network connections
Rapid Spanning Tree Protocol (RSTP)	Prevent Layer 2 loops	Fast convergence
Link Aggregation Control Protocol (LACP)	Bundle multiple uplinks	Increased bandwidth & redundancy
Port Security	Limit MAC addresses	Prevent unauthorized devices
Bridge Protocol Data Unit (BPDU) Guard	Protect STP topology	Stops rogue switches
Storm Control	Prevent broadcast storms	Improves stability
QoS	Prioritize voice/video	Better application performance
Power over Ethernet PoE+	Power phones/APs	Simplifies deployment

Networking Appliances, Applications & Functions

Enterprise networking appliances perform specialized but complementary roles across the OSI model. Some provide connectivity and routing, others deliver centralized services, and still others protect or optimize network traffic. Understanding where each operates and how its function differs from related technologies helps determine where it belongs within an enterprise design.

Appliance	Function	OSI Layer	Enterprise Role / Example
Layer 2 Switch	Connects devices within a LAN or VLAN	L2	Access switch connecting PCs, phones, printers, and APs on an office floor
Wireless Access Point (WAP)	Provides wireless network connectivity	L2	Connects wireless employees and devices in offices and conference rooms
Router	Connects different networks and forwards packets	L3	Routes traffic between the enterprise network, WAN, or external networks
Layer 3 Switch	Combines switching with Layer 3 routing	L2–L3	Provides high-speed inter-VLAN routing within the MDF/core
VPN Gateway	Establishes encrypted VPN connections	L3+	Provides secure remote-access or site-to-site connectivity
Firewall / NGFW	Filters and inspects traffic according to security policy	L3–L7	Protects the enterprise at the Internet edge and controls traffic between security zones
Intrusion Detection System (IDS)	Detects and alerts on suspicious or malicious activity	L3–L7	Monitors network traffic and generates security alerts without automatically blocking it
Intrusion Prevention System (IPS)	Detects and actively blocks malicious activity	L3–L7	Operates inline to prevent identified threats from reaching protected systems
Load Balancer	Distributes client requests across multiple servers	L4 / L7	Improves application performance, availability, and fault tolerance
DHCP Server	Automatically provides IP configuration	L7	Assigns IP addresses and related network settings to client devices

Appliance	Function	OSI Layer	Enterprise Role / Example
DNS Server	Resolves hostnames to IP addresses	L7	Allows users and applications to locate internal and external network resources

VLAN Segmentation, IP Addressing & Network Services

The proposed network uses Virtual Local Area Networks (VLANs) to logically separate business units, servers, and guest devices while allowing them to share the same physical switching infrastructure. Each VLAN forms an independent broadcast domain with its own IPv4 subnet and IPv6 /64 prefix, reducing unnecessary broadcast traffic while improving performance, security, manageability, and troubleshooting. Private IPv4 addressing from the 10.0.0.0/8 address space provides room for organizational growth, while a dual-stack IPv4/IPv6 implementation maintains compatibility with current systems and prepares the network for future requirements.

IEEE 802.1Q trunking carries multiple VLANs between switches, while Layer 3 switching provides inter-VLAN routing when communication between VLANs is required. Access Control Lists (ACLs) restrict this communication according to business and security requirements. Centralized network services provide addressing, name resolution, time synchronization, authentication, and secure remote access, creating a structured network that is easier to secure, manage, scale, and troubleshoot.

VLAN / Function	IPv4 Network	Example IPv6 Prefix	Addressing	Enterprise Access
VLAN 10 – IT, Security & Operations	10.10.0.0/16	2001:db8:100::/64	DHCP / IPv6 auto-configuration	DNS, AD, internal resources
VLAN 11 – Business A	10.20.0.0/16	2001:db8:200::/64	DHCP / IPv6 auto-configuration	DNS, AD, authorized resources
VLAN 12 – Business B	10.30.0.0/16	2001:db8:300::/64	DHCP / IPv6 auto-configuration	DNS, AD, authorized resources
VLAN 13 – Business C	10.40.0.0/16	2001:db8:400::/64	DHCP / IPv6 auto-configuration	DNS, AD, authorized resources

VLAN / Function	IPv4 Network	Example IPv6 Prefix	Addressing	Enterprise Access
VLAN 17 – Servers	10.50.0.0/16	2001:db8:500::/64	Static	Hosts centralized services
VLAN 19 – Guest Wi-Fi	10.99.0.0/16	2001:db8:990::/64	DHCP / IPv6 auto-configuration	Internet only; isolated from internal LAN

The IPv6 addresses shown use the IETF documentation prefix `2001:db8::/32` to illustrate the addressing plan. Production IPv6 prefixes would be based on addresses assigned to the organization.

Centralized Network Services

Service	Runs On	Used By	Purpose
DHCP	DHCP Server	IPv4 clients	Automatic IPv4 addressing
DHCPv6 / SLAAC	Router or DHCPv6 Server	IPv6 clients	IPv6 address configuration
DNS	DNS Server	Network users/devices	Hostname resolution
NTP	Time Server	Network devices and systems	Time synchronization
Active Directory	Domain Controller	Employees and managed systems	Centralized authentication and authorization
VPN	VPN Gateway / Firewall	Remote employees	Secure remote connectivity

Cloud Concepts & Connectivity

Cloud computing extends enterprise capabilities beyond the traditional data center by providing scalable, flexible, and highly available services. The proposed network uses a hybrid cloud model, keeping critical services such as Active Directory, DNS, DHCP, and file services on-premises while using cloud resources for Microsoft 365, SaaS applications, backups, disaster recovery, and other scalable services. This approach balances organizational control, security, performance, cost, and future growth.

Cloud Deployment & Infrastructure Concepts

Cloud Concept	Enterprise Use	Key Consideration
Public Cloud	Microsoft 365, backups, collaboration, web services	Scalable and cost-efficient, but provides less infrastructure control
Private Cloud	Dedicated applications and sensitive workloads	Greater control and customization, but higher cost and management requirements
Hybrid Cloud	Combines on-premises infrastructure with cloud services	Proposed model; balances control, flexibility, scalability, and cost
Virtualization	Runs multiple virtual systems on shared physical hardware	Improves resource utilization, deployment flexibility, and disaster recovery
Elasticity	Resources expand or contract with changing demand	Matches resources to current workload requirements
Scalability	Capacity increases as organizational needs grow	Supports additional users, applications, and services over time
CDN	Distributes content across geographically dispersed servers	Reduces latency and improves availability

Cloud Connectivity

Cloud connectivity securely links users, enterprise networks, and cloud resources. The appropriate method depends on requirements for security, bandwidth, latency, cost, and scalability.

Connectivity Option	Enterprise Use
Remote-Access VPN	Provides encrypted network access for individual remote employees
Site-to-Site VPN	Securely connects enterprise networks and cloud resources across the Internet; *recommended for the proposed design*
Dedicated Cloud Connection	Provides dedicated, high-bandwidth, low-latency cloud connectivity for larger workloads

Connectivity Option	Enterprise Use
Colocation	Places organization-owned equipment in a third-party data center
Transit Gateway	Centralizes connectivity among multiple cloud networks and VPN connections

Cloud Service Models

Cloud service models differ primarily in how management responsibilities are divided between the provider and the organization. SaaS requires the least infrastructure management, PaaS provides a managed application-development platform, and IaaS provides greater control over virtual infrastructure.

Service Model	Enterprise Example	What It Provides	Primary Advantage
SaaS	Microsoft 365	Complete applications delivered as a service	Lowest management overhead and rapid deployment
PaaS	Azure App Service	Managed platform for building and deploying applications	Simplifies application development and deployment
IaaS	Azure Virtual Machines	Virtual servers, storage, and networking resources	Greater infrastructure control and flexibility

Proposed Approach

A hybrid cloud model provides the best fit for the proposed enterprise because it preserves local control of critical network services while allowing cloud resources to provide scalability, remote accessibility, backup, disaster recovery, and application services. Secure VPN connectivity integrates these environments while maintaining protected communication between users, organizational networks, and cloud resources.

Security Considerations

A secure enterprise network relies on defense-in-depth, where multiple administrative, physical, and technical security controls work together to protect organizational resources. Rather than relying on a single security device, the proposed design implements complementary controls that prevent unauthorized access, protect critical systems, support business continuity, and improve the organization's ability to detect and respond to cyber threats.

Security Layer	Recommended Controls	Enterprise Benefit
Physical Security	Secure MDF/IDF rooms, environmental controls, access logging	Protects critical network infrastructure from unauthorized physical access
Identity & Access Management	Active Directory, MFA, IEEE 802.1X, Principle of Least Privilege (PoLP)	Verifies user and device identities while limiting unauthorized access
Network Security	Next-Generation Firewalls (NGFW), IDS/IPS, VLANs, ACLs, encrypted VPNs	Protects network traffic, isolates departments, and secures remote connectivity
System Protection	Patch management, backups, disaster recovery, security awareness training	Reduces vulnerabilities and supports business continuity
Monitoring & Response	SNMP, Wireshark, Nmap, SIEM, Endpoint Detection & Response (EDR)	Detects, investigates, and responds to security and network events

Together, these layered security controls complement the proposed hierarchical architecture, VLAN segmentation, centralized network services, and redundant infrastructure to provide a secure, resilient, and manageable enterprise network. This defense-in-depth approach protects organizational resources while supporting business continuity, regulatory best practices, and future organizational growth.

Conclusion

The proposed enterprise network architecture provides a secure, reliable, scalable, and highly available infrastructure designed to support the organization's current operations while remaining flexible for future growth. By integrating a hierarchical network design with enterprise switching technologies, VLAN segmentation, redundant switching, cloud connectivity, centralized network services, IPv4 and IPv6 addressing, network appliances, and layered defense-in-depth security controls, the design delivers efficient communication, simplified management, resilient connectivity, and improved overall performance.

Redundancy and segmentation reduce single points of failure, strengthen security, and support business continuity, resulting in a modern enterprise network that is both maintainable and adaptable to evolving business and technology requirements. Ultimately, successful network design extends beyond selecting hardware or configuring devices; it begins with understanding business objectives and applying industry standards, sound engineering principles, and proven networking practices to build an infrastructure capable of supporting the organization's long-term success.

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